ReneWind

Intro Neural Networks: AI/ML

Date: February 2026

Karen Wood

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Contents / Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- Data Overview
- EDA Results
- Data Preprocessing
- Model Evaluation Criterion
- Model Performance Summary
- Test Performance of Best Model

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Executive Summary

Business Objective

Reduce wind turbine downtime and maintenance costs by predicting generator failures early, enabling repairs before costly breakdowns occur.

Key Challenge

Missed failures lead to expensive replacements, while false alarms only trigger inspections. Therefore, the solution prioritizes maximizing failure detection (recall) over overall accuracy.

Approach

Sensor-based machine learning models were developed to identify turbines at risk of failure, with a focus on maximizing early detection.

Outcome

The final model detects approximately 90% of generator failures while balancing inspection volume through threshold tuning. ~90% recall on the test set at threshold = 0.35.

Business Impact

By shifting maintenance from reactive replacement to proactive repair, ReneWind can lower operational costs and improve turbine availability.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Problem Overview and Solution Approach

Business Problem

- Wind turbine generator failures are costly and disruptive.
  - Missed failures result in emergency replacements and extended downtime
  - False alarms only require inspections, which are significantly cheaper
- Because the cost of missing a failure far exceeds the cost of inspection, the business requires a solution that prioritizes early failure detection.

Solution Approach

- Collected and analyzed anonymized sensor data from wind turbines
- Built multiple classification models to predict generator failure
- Selected models based on failure recall, not accuracy
- Tuned probability thresholds to align model decisions with business cost tradeoffs

The final model is designed to act as an early-warning system, flagging turbines for inspection or repair before breakdown occurs.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

Target Distribution

- Generator failures (Target = 1) represent a small minority of observations
- The dataset is highly imbalanced, making accuracy an unreliable evaluation metric

Feature Characteristics

- All predictors are numeric, anonymized, sensor-derived features
- No categorical variables or timestamps are present
- Sensor values vary widely in scale and distribution

Missing Values

- A small number of missing values were observed in a limited set of variables
- Missingness was minimal and addressed using median imputation

Correlation Insights

- Individual sensor features exhibit low to moderate pairwise correlation
- No single feature strongly explains generator failures on its own
- This suggests failures are driven by complex, multi-sensor interactions

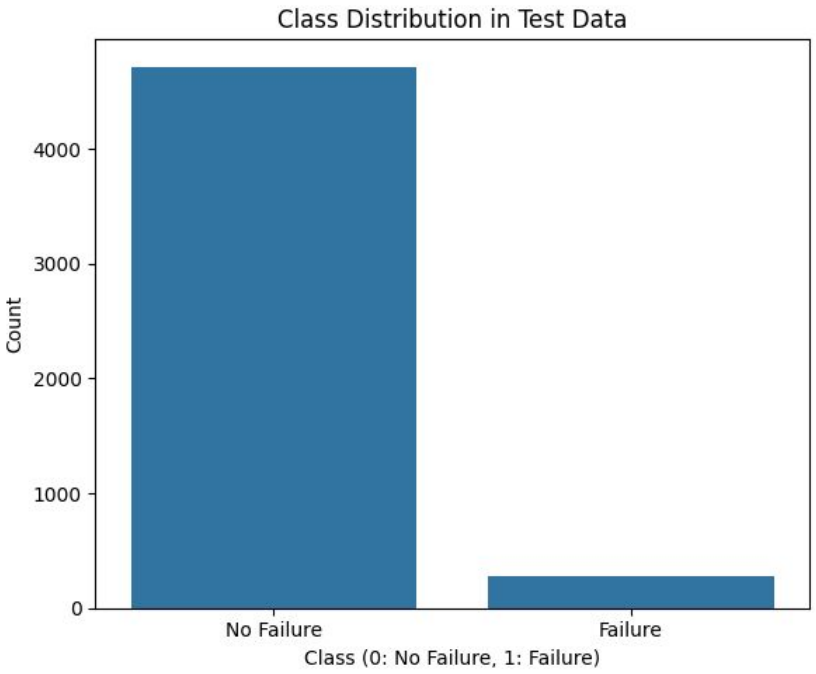
Key Implications for Modeling

- Class imbalance and weak individual feature signals require:
    - Models capable of capturing non-linear relationships
    - Evaluation metrics focused on failure recall rather than accuracy

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results: Univariate Analysis – Target Distribution

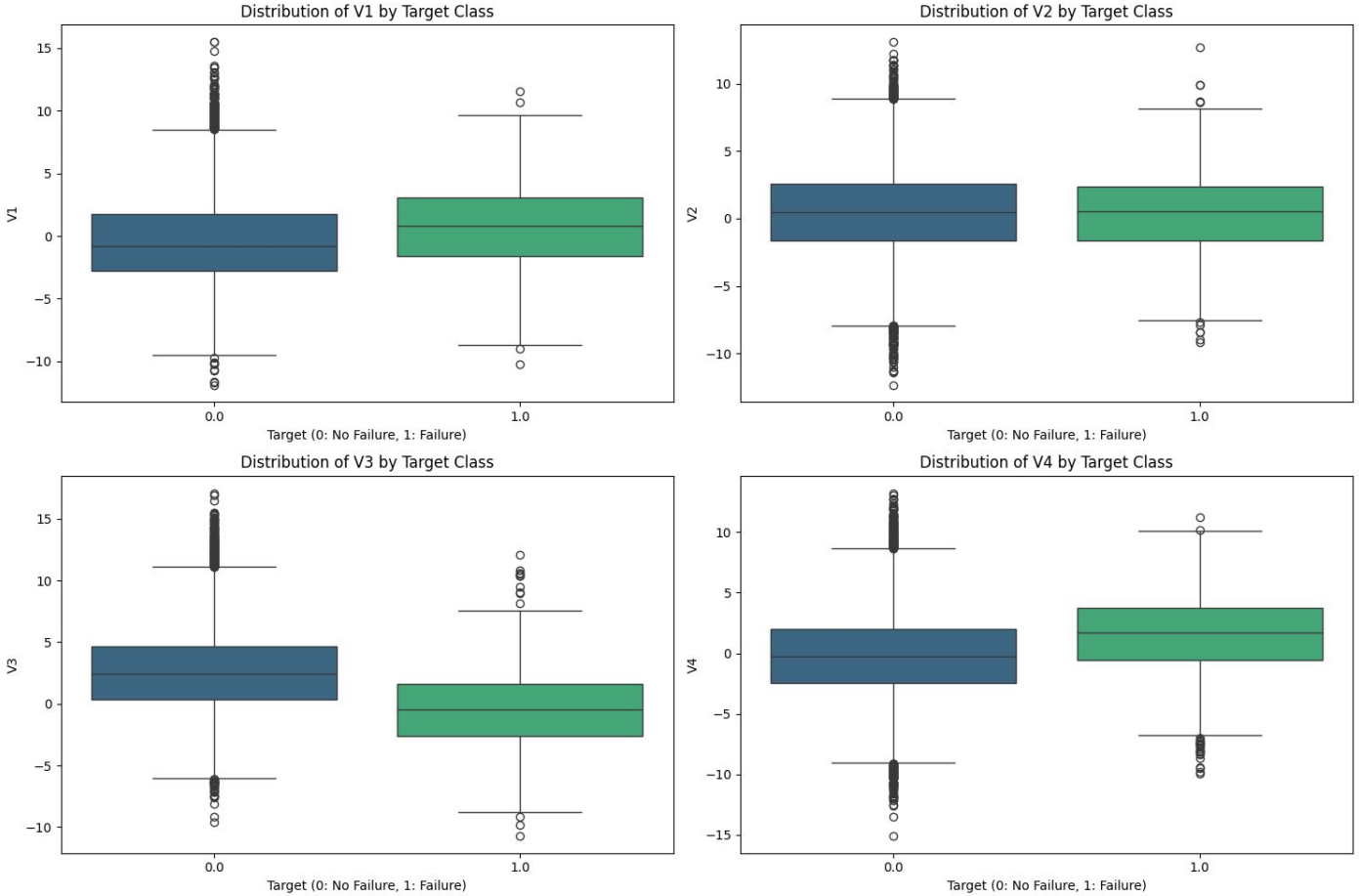
Generator failures represent a small minority of observations, confirming severe class imbalance



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results: Bivariate Analysis – Feature vs Failure Outcome

- Bivariate analysis compares individual sensor features against the failure outcome.
- Representative sensor features selected to illustrate overlapping distributions between failure and non-failure cases.
- No single sensor strongly separates failures.



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

Duplicate check
No duplicate records were found in the training dataset.

Missing value treatment
A small number of missing values were identified in select variables and handled using median imputation to preserve distribution shape and avoid outlier influence.

Outlier assessment
Sensor values showed natural variability consistent with operational data; no aggressive outlier removal was applied to avoid loss of failure-related signals.

Train–validation split
The training data was split into training and validation sets using stratified sampling to preserve the failure class distribution.

Data preparation for modeling
Features and target variables were separated, and all preprocessing steps were applied consistently across training, validation, and test datasets to prevent data leakage.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Evaluation Criterion

Primary Metric: Recall (Failure Class)

- The primary objective is to detect generator failures early
- False negatives (missed failures) result in costly generator replacement and downtime
- False positives only trigger inspections, which are comparatively low cost

Rationale

- The dataset is highly imbalanced, making accuracy misleading
- Maximizing recall for the failure class (Target = 1) minimizes missed failures
- Precision is considered secondarily and managed through threshold tuning

Evaluation Strategy

- Models were compared using failure recall on the validation set
- Probability thresholds were tuned to balance failure detection and inspection volume
- The final model was evaluated once on the test set to estimate real-world performance

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Neural Network Experimentation (≥6 Model Variants)

Models vary systematically across depth, optimizer, dropout, and class weighting to improve recall and generalization.

Model	Hidden layers	Units (incl. output)	Optimizer	Dropout	Class weights
model_0	1	[7, 1]	SGD	No	No
model_1	2	[16, 16, 1]	SGD	No	No
model_2	3	[16, 16, 8, 1]	SGD	Yes	No
model_3	3	[16, 16, 8, 1]	SGD	Yes	Yes
model_4	2	[16, 16, 1]	Adam	No	No
model_5	3	[32, 16, 8, 1]	Adam	Yes	No
model_6	3	[32, 16, 8, 1]	SGD	Yes	Yes

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary

Model Comparison (Training & Validation)

- Seven neural network models (Models 0–6) were evaluated
- Models varied by network depth, regularization (dropout), optimizer choice, and use of class weighting
- Performance was compared using failure recall as the primary metric

Key Observations

- Simpler models (Models 0–2) showed limited ability to capture failure patterns
- Adding regularization and depth improved generalization performance
- Models incorporating class weighting consistently achieved higher failure recall
- Model 6 demonstrated the strongest validation recall with stable performance across datasets

Final Model Selection

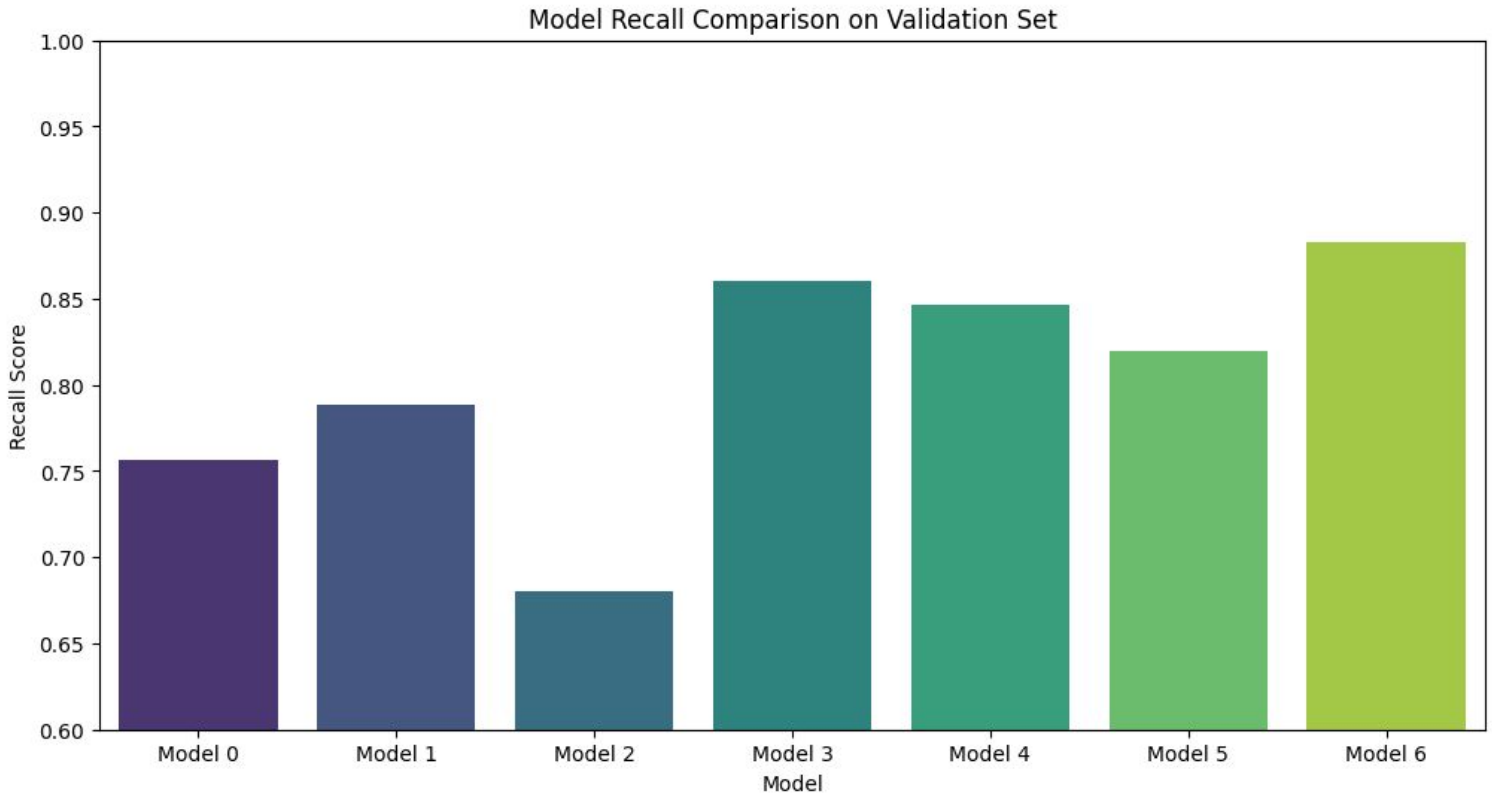
- Model 6 was selected as the final model based on:
  - Highest recall for the failure class on the validation set
  - Effective handling of class imbalance
  - Consistent performance without signs of overfitting

	Model 0	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Accuracy	0.982250	0.985500	0.978750	0.969750	0.989500	0.989000	0.971750
Recall	0.756757	0.788288	0.680180	0.860360	0.846847	0.819820	0.882883
Precision	0.908108	0.940860	0.915152	0.679715	0.959184	0.978495	0.692580
F1 Score	0.825553	0.857843	0.780362	0.759443	0.899522	0.892157	0.776238

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Comparison (Validation Set)

Recall was chosen as the primary metric because the cost of missed generator failures far exceeds the cost of unnecessary inspections; accuracy and F1-score treat both error types equally and are therefore misaligned with the business objective.

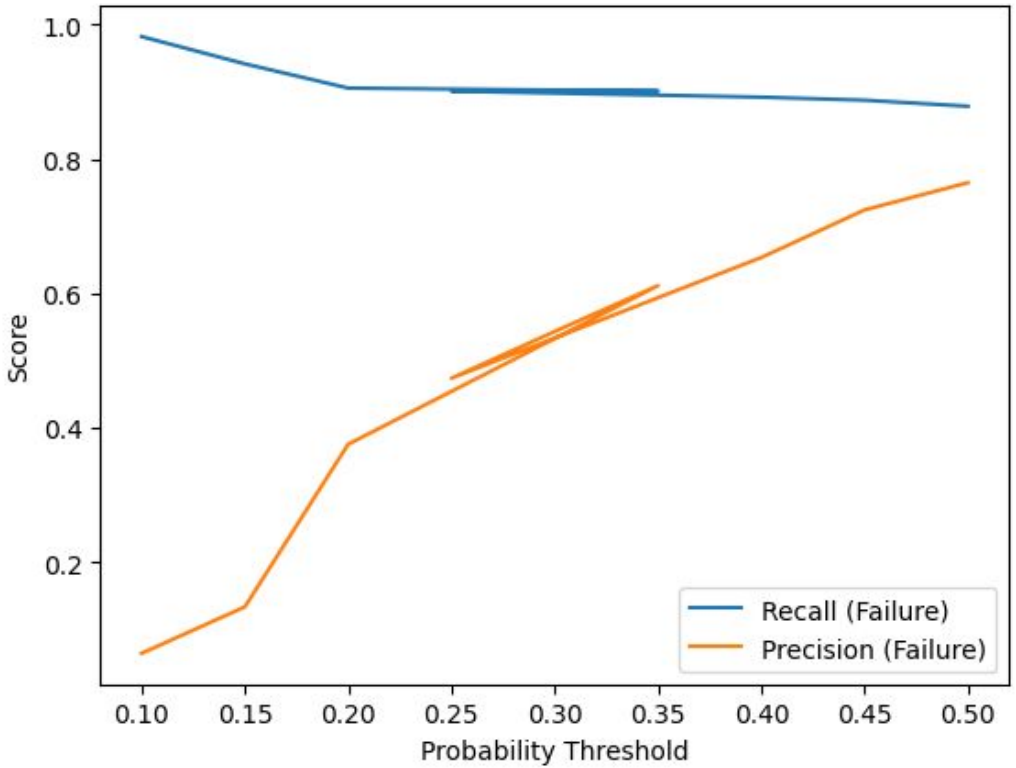


Test Confusion Matrix

- False negatives = 40 (missed failures)
- False positives = 110 (inspections)

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Threshold Tuning



- Threshold tuning allows us to control the tradeoff between early failure detection and inspection costs; a threshold of 0.35 achieves high failure recall while substantially reducing false alarms.
- Selected operating threshold = 0.35 to maximize recall while limiting inspection volume.
- Between thresholds 0.25 and 0.35, failure recall remains stable at ~90%, while precision improves significantly.
- ReneWind can reduce unnecessary inspections without increasing missed failures.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Best Model - Test Performance

Final Model Description

- The final model (6) is a more advanced version of the initial baseline model
- It uses multiple layers to learn complex patterns from sensor data
- Dropout was added to reduce overfitting and improve performance on new data
- The model was trained to give extra importance to failure cases, since failures are rare but costly

Performance on Test Data

- Model 6 was evaluated on a separate test dataset that was not used during training or tuning
- This evaluation represents how the model would perform on new, real-world turbines
- The model successfully identified most generator failures in the test data
- Adjusting the probability threshold helped balance:
  - catching failures early
  - avoiding excessive inspections

Business Interpretation

- Model 6 acts as an early warning system for maintenance teams
- Turbines with a high risk of failure can be inspected or repaired before breakdown occurs thus reducing unexpected downtime, emergency generator replacements and overall maintenance costs

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Final Recommendations

Recommended Model

- Deploy Neural Network Model 6 as the production failure-prediction model

Operating Threshold

- Use a probability threshold of 0.35 to trigger inspections

Rationale

- Achieves ~90% recall on the test set, minimizing missed generator failures
- Effectively handles severe class imbalance through class weighting and regularization
- Provides a strong tradeoff between early failure detection and inspection volume

Business Action

- Use model outputs as an early-warning system
- Flag high-risk turbines for preventive inspection or repair before breakdown
- Shift maintenance strategy from reactive replacement to proactive cost-efficient intervention

Expected Impact

- Fewer emergency generator replacements
- Reduced turbine downtime
- Lower overall maintenance costs with manageable inspection overhead

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.